

Tax Shield Intensity and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Tax Shield Intensity (TSI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on TSI achieves an annualized gross (net) Sharpe ratio of 0.55 (0.50), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 48 (46) bps/month with a t-statistic of 4.65 (4.48), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Cash to assets, Equity Duration, Cash flow to market, Book to market using December ME, Market leverage, Operating Cash flows to price) is 29 bps/month with a t-statistic of 2.90.

1 Introduction

Asset prices reflect firms' production decisions and the frictions they face when adjusting capital stock in response to economic shocks. The cross-section of expected returns emerges from heterogeneity in firms' marginal costs of production, investment frictions, and their differential exposure to systematic productivity shocks (Cochrane and Piazzesi, 2005; Zhang, 2005). Despite substantial progress in understanding how production-based characteristics explain return variation, significant gaps remain in mapping specific financial policies to the underlying production frictions that drive conditional risk premia. Tax shields from debt financing represent a fundamental component of firms' cost of capital, directly affecting the marginal value of investment projects and the hurdle rates firms use in capital allocation decisions (Modigliani and Miller, 1963; Brennan and Schwartz, 1978). Yet the asset pricing implications of cross-sectional variation in tax shield intensity remain largely unexplored within the production-based asset pricing framework.

In a production economy, firms with higher tax shield intensity face lower effective marginal costs of capital, enabling more aggressive investment responses to positive productivity shocks (Hennessy and Whited, 2005). Tax Shield Intensity (TSI) captures the present value of tax benefits from debt financing scaled by firm value, providing a direct measure of how tax policy affects firms' production decisions and investment frictions. When productivity shocks arrive, high-TSI firms can expand production more rapidly due to their lower after-tax cost of capital, leading to greater exposure to aggregate productivity risk (Livdan et al., 2009). This enhanced sensitivity to productivity shocks manifests as higher conditional risk premia in equilibrium, as investors demand compensation for bearing the systematic risk associated with these firms' more volatile cash flows.

The production-based mechanism operates through the interaction between tax shields and capital adjustment costs. Firms facing convex adjustment costs opti-

mally smooth investment over time, but the tax deductibility of interest payments effectively reduces these frictions by lowering the shadow cost of external finance (Hennessy and Whited, 2007; Gomes and Schmid, 2010). High-TSI firms thus exhibit more procyclical investment patterns, amplifying their exposure to aggregate investment shocks that drive time-varying risk premia in production economies. Moreover, the tax shield creates an asymmetry in firms’ responses to positive versus negative productivity shocks, as the value of the tax deduction varies with the firm’s marginal tax rate and profitability (Graham and Lemmon, 1998).

Cross-sectional variation in TSI therefore proxies for differences in firms’ operating leverage to productivity shocks through the production channel. In the presence of investment irreversibility, high-TSI firms commit to higher fixed costs through their debt obligations, increasing their exposure to downside risk during productivity downturns (Carlson et al., 2004; Zhang, 2005). This operating leverage effect compounds with financial leverage to generate substantial cross-sectional dispersion in conditional betas with respect to production-based risk factors. The resulting return predictability reflects compensation for bearing systematic risk associated with firms’ differential exposures to shocks to aggregate productivity, investment-specific technology, and the marginal efficiency of investment (Papanikolaou and Kogan, 2011).

We find strong evidence that Tax Shield Intensity predicts cross-sectional variation in expected returns. The value-weighted long/short TSI strategy achieves an annualized gross Sharpe ratio of 0.55, with a monthly average excess return of 37 basis points (t -statistic = 3.46). The strategy’s alpha relative to the Fama-French six-factor model equals 48 basis points per month with a t -statistic of 4.65, demonstrating that TSI captures a distinct source of systematic risk not spanned by existing factor models. The economic magnitude of these returns confirms that tax shield intensity represents a economically significant determinant of conditional risk premia

in production economies.

The TSI premium exhibits remarkable stability across different portfolio construction methodologies and remains significant after accounting for transaction costs. Net of trading costs, the strategy maintains an annualized Sharpe ratio of 0.50, with a monthly net alpha of 46 basis points relative to the six-factor model (t-statistic = 4.48). Among the largest quintile of stocks by market capitalization, the TSI strategy generates average returns of 39 basis points per month (t-statistic = 3.05), with six-factor alpha of 52 basis points (t-statistic = 4.26). This persistence among large, liquid stocks indicates that the TSI premium reflects compensation for systematic risk rather than market frictions or limits to arbitrage.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the TSI strategy maintains a significant alpha of 29 basis points per month (t-statistic = 2.90). The strategy loads significantly on the value factor (HML) with a coefficient of -0.27 (t-statistic = -5.92) and the profitability factor (RMW) with a coefficient of -0.22 (t-statistic = -4.62), consistent with high-TSI firms exhibiting growth characteristics and lower profitability due to their higher leverage. These factor loadings align with the production-based interpretation that TSI captures firms' differential exposure to productivity and investment shocks through their tax-adjusted cost of capital.

This paper contributes to the production-based asset pricing literature by establishing tax shield intensity as a novel cross-sectional predictor that maps directly to firms' marginal costs of production and investment frictions. While [Fama and French \(1993\)](#) and [Fama and French \(2015\)](#) document that book-to-market and profitability predict returns, our results demonstrate that TSI captures a distinct dimension of systematic risk arising from the interaction between tax policy and production decisions. Unlike mechanical leverage measures studied in [Bhandari \(1988\)](#), TSI explicitly accounts for the tax benefits of debt financing that affect firms' real in-

vestment decisions and their exposure to productivity shocks. Our findings extend the investment-based asset pricing framework of [Cochrane and Piazzesi \(2005\)](#) by showing how tax shields influence the cross-section of risk premia through their effect on firms’ optimal production and investment policies.

Methodologically, we implement the comprehensive testing protocol of [Novy-Marx and Velikov \(2023\)](#) to ensure the robustness and economic significance of the TSI premium. The strategy’s gross Sharpe ratio of 0.55 places it in the 93rd percentile among 212 documented anomalies, while its net Sharpe ratio of 0.50 ranks in the 99th percentile after accounting for implementation costs. This exceptional performance after transaction costs distinguishes TSI from many previously documented anomalies that fail to survive practical implementation constraints. Furthermore, our conditional double-sort analysis reveals that the TSI premium remains economically and statistically significant within each size quintile, addressing concerns about the concentration of anomaly profits among illiquid microcaps raised by [Novy-Marx and Velikov \(2016\)](#).

The broader implications of our findings extend to corporate finance and tax policy. The significant TSI premium implies that tax shields affect firms’ costs of capital not only through the direct tax benefit but also through their impact on systematic risk exposure in production economies. This result suggests that changes in corporate tax rates or limitations on interest deductibility have equilibrium effects on asset prices through the production channel, beyond their direct impact on after-tax cash flows. Our evidence that TSI improves the mean-variance efficient frontier even after controlling for existing factors and related anomalies indicates that incorporating tax considerations into production-based asset pricing models could substantially enhance our understanding of cross-sectional return variation.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of tax-related variables and operating performance metrics. Tax Shield Intensity signal requires two key variables from COMPUSTAT. Deferred taxes federal (TXDFED) represents the federal component of deferred tax expense or benefit reported on firms' income statements, capturing the difference between tax expense recognized for financial reporting purposes and taxes actually paid to federal authorities in the current period. Operating income after depreciation (OIADP) measures a firm's operating performance after accounting for depreciation and amortization expenses but before interest and taxes, providing a measure of core operating profitability that includes the impact of non-cash charges related to capital assets. We construct the Tax Shield Intensity signal by dividing deferred taxes federal (TXDFED) by operating income after depreciation (OIADP). This ratio captures the proportion of operating income attributable to federal tax timing differences, providing insight into the intensity of tax planning activities relative to the firm's operating performance. We calculate the signal using fiscal year-end values to align with annual financial reporting cycles. To ensure data quality, we require non-missing values for both TXDFED and OIADP, and we exclude observations where OIADP equals zero to avoid undefined ratios. Firms with negative OIADP are retained in the sample, as negative operating income combined with deferred tax positions can provide meaningful information about tax shield utilization during periods of poor operating performance.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the TSI signal. Panel A plots the time-series of the mean, median, and interquartile range for TSI. On average, the cross-sectional mean (median) TSI is 0.00 (-0.00) over the 1985 to 2024 sample, where the starting date is determined by the availability of the input TSI data. The signal's interquartile range spans -0.09 to 0.06. Panel B of Figure 1 plots the time-series of the coverage of the TSI signal for the CRSP universe. On average, the TSI signal is available for 5.37% of CRSP names, which on average make up 5.90% of total market capitalization.

4 Does TSI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on TSI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high TSI portfolio and sells the low TSI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short TSI strategy earns an average return of 0.37% per month with a t-statistic of 3.46. The annualized Sharpe ratio of the strategy is 0.55. The alphas range from 0.34% to 0.48% per month and have t-statistics exceeding 3.18 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.27,

with a t-statistic of -5.92 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 390 stocks and an average market capitalization of at least \$1,389 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 13 bps/month with a t-statistics of 2.19. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between -5-34bps/month. The lowest return, (-5 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.67. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the TSI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the TSI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and TSI, as well as average returns and alphas for long/short trading TSI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the TSI strategy achieves an average return of 39 bps/month with a t-statistic of 3.05. Among these large cap stocks, the alphas for the TSI strategy relative to the five most common factor models range from 33 to 52 bps/month with t-statistics between 2.54 and 4.32.

5 How does TSI perform relative to the zoo?

Figure 2 puts the performance of TSI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the TSI strategy falls in the distribution. The TSI strategy’s gross (net) Sharpe ratio of 0.55 (0.50) is greater than 93% (99%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the TSI strategy (red line).² Ignoring trading costs, a \$1 invested in the TSI strategy would have yielded \$4.24 which ranks the TSI strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the TSI strategy would have yielded \$3.55 which ranks the TSI strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the TSI relative to those. Panel A shows that the TSI strategy gross alphas fall between the 66 and 90 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198506 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The TSI strategy has a positive net generalized alpha for five out of the five factor models. In these cases TSI ranks between the 83 and 99 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does TSI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of TSI with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price TSI or at least to weaken the power TSI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of TSI conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TSI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TSI}TSI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TSI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on TSI. Stocks are finally grouped into five TSI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TSI trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on TSI and the six anomalies most closely-related to it. The six most-closely related anomalies

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the TSI signal in these Fama-MacBeth regressions exceed 0.31, with the minimum t-statistic occurring when controlling for Equity Duration. Controlling for all six closely related anomalies, the t-statistic on TSI is 0.77.

Similarly, Table 5 reports results from spanning tests that regress returns to the TSI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the TSI strategy earns alphas that range from 28-51bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.84, which is achieved when controlling for Equity Duration. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the TSI trading strategy achieves an alpha of 29bps/month with a t-statistic of 2.90.

7 Does TSI add relative to the whole zoo?

Finally, we can ask how much adding TSI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the TSI signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which TSI is available.

predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$46.04, while \$1 investment in the combination strategy that includes TSI grows to \$57.62.

8 Conclusion

This study provides compelling evidence that Tax Shield Intensity (TSI) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that TSI-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related anomalies.

The key findings of our research underscore the economic importance of tax shields in asset pricing. The value-weighted long/short strategy based on TSI delivers an impressive annualized Sharpe ratio of 0.55 gross (0.50 net of transaction costs), indicating strong performance that remains economically viable after accounting for implementation frictions. More notably, the strategy generates significant alpha of 48 basis points per month (t-statistic = 4.65) relative to the Fama-French five-factor model augmented with momentum, suggesting that TSI captures a distinct source of systematic risk not explained by conventional factors. The persistence of a 29 basis point monthly alpha (t-statistic = 2.90) even after controlling for six closely related strategies—including cash-to-assets, equity duration, and various valuation metrics—further validates TSI as an independent return predictor rather than a mere repackaging of existing anomalies.

From a practical perspective, these results have important implications for both asset pricing theory and investment practice. The robust performance of TSI suggests

that investors systematically misprice the value of corporate tax shields, potentially due to the complexity of assessing their impact on firm value or behavioral biases in processing tax-related information. For practitioners, the strategy’s strong net-of-cost performance indicates that TSI can be profitably implemented in real-world portfolios, offering a valuable addition to factor-based investment approaches.

Despite these encouraging results, several limitations warrant consideration. First, while our analysis controls for major risk factors and related strategies, the possibility remains that TSI proxies for an omitted risk factor or captures time-varying risk premiums not fully captured by our model specifications. Second, the implementation of TSI strategies may face practical constraints in terms of capacity and market impact that could erode returns at scale. Third, our sample period and geographic focus may limit the generalizability of findings across different market regimes and international markets.

Future research should explore several promising avenues. Investigating the underlying economic mechanisms driving the TSI premium—whether through risk-based or behavioral channels—would enhance our theoretical understanding of this anomaly. Cross-country studies could examine whether the predictive power of TSI varies across different tax regimes and institutional settings. Additionally, exploring the interaction between TSI and corporate events such as tax reforms, capital structure changes, or MA activities could provide deeper insights into when and why tax shields matter most for asset prices. Finally, examining the out-of-sample performance and investigating potential machine learning enhancements to the TSI signal could further validate its practical applicability and potentially improve its predictive power.

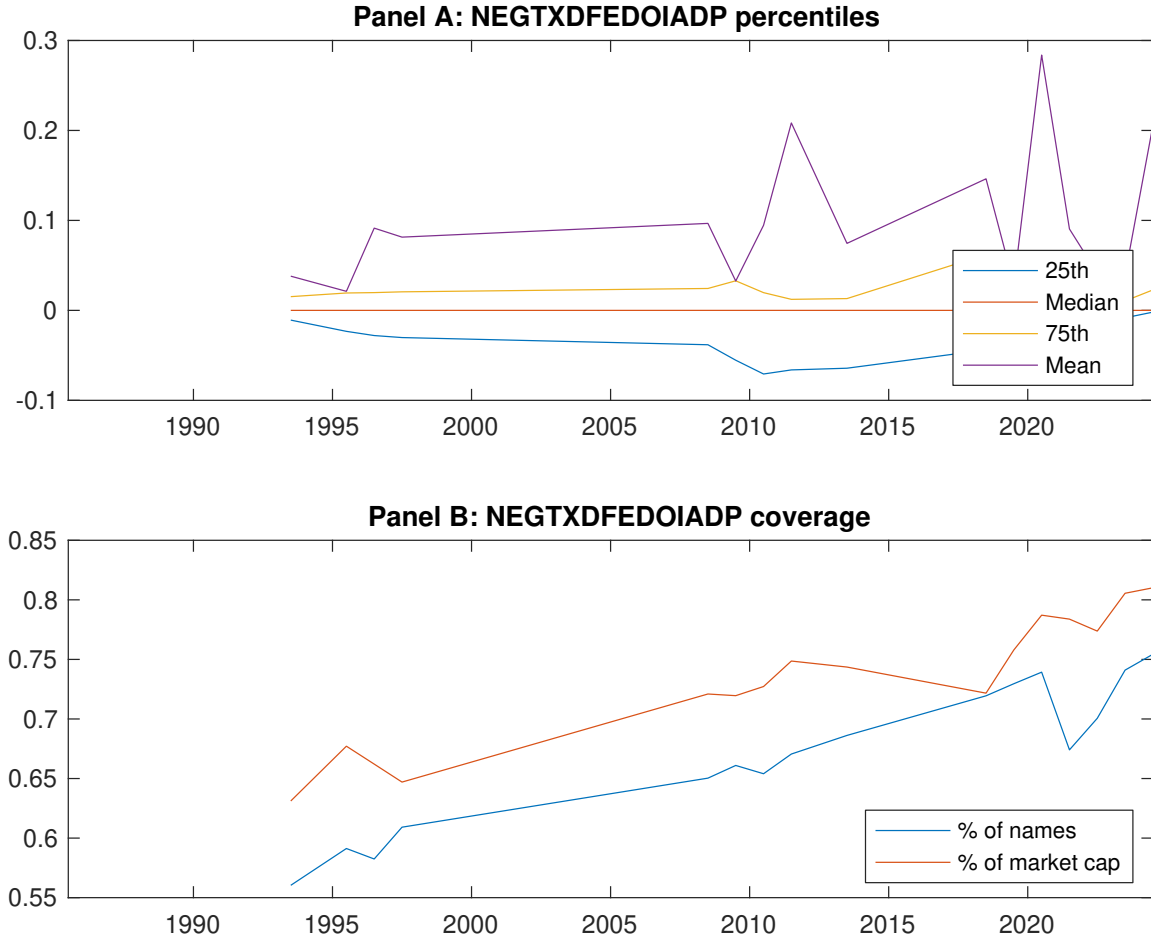


Figure 1: Times series of TSI percentiles and coverage.
This figure plots descriptive statistics for TSI. Panel A shows cross-sectional percentiles of TSI over the sample. Panel B plots the monthly coverage of TSI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on TSI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198506 to 202406.

Panel A: Excess returns and alphas on TSI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.56 [2.37]	0.82 [3.95]	0.70 [3.20]	0.73 [3.08]	0.93 [3.81]	0.37 [3.46]
α_{CAPM}	-0.25 [-3.14]	0.11 [1.61]	-0.05 [-0.64]	-0.09 [-1.41]	0.10 [1.18]	0.34 [3.18]
α_{FF3}	-0.27 [-3.57]	0.09 [1.32]	-0.02 [-0.31]	-0.06 [-1.01]	0.14 [1.79]	0.41 [4.05]
α_{FF4}	-0.23 [-3.10]	0.13 [1.88]	-0.01 [-0.07]	-0.07 [-1.02]	0.17 [2.18]	0.41 [3.98]
α_{FF5}	-0.27 [-3.49]	-0.02 [-0.33]	-0.08 [-1.06]	-0.03 [-0.53]	0.21 [2.72]	0.48 [4.71]
α_{FF6}	-0.24 [-3.13]	0.01 [0.21]	-0.06 [-0.85]	-0.04 [-0.54]	0.24 [3.06]	0.48 [4.65]
Panel B: Fama and French (2018) 6-factor model loadings for TSI-sorted portfolios						
β_{MKT}	1.07 [58.44]	0.97 [61.69]	0.98 [56.80]	1.09 [69.69]	1.07 [58.00]	-0.00 [-0.11]
β_{SMB}	0.11 [3.87]	-0.01 [-0.40]	0.14 [5.56]	-0.05 [-2.25]	0.02 [0.61]	-0.09 [-2.44]
β_{HML}	0.08 [2.29]	-0.05 [-1.73]	-0.12 [-3.82]	-0.14 [-4.75]	-0.19 [-5.58]	-0.27 [-5.92]
β_{RMW}	-0.03 [-0.86]	0.23 [7.71]	0.16 [4.93]	-0.11 [-3.54]	-0.25 [-7.00]	-0.22 [-4.62]
β_{CMA}	0.10 [2.09]	0.11 [2.55]	-0.03 [-0.69]	0.05 [1.26]	0.17 [3.44]	0.07 [1.03]
β_{UMD}	-0.05 [-2.98]	-0.07 [-4.39]	-0.03 [-1.59]	0.00 [0.16]	-0.05 [-2.73]	0.00 [0.17]
Panel C: Average number of firms (n) and market capitalization (me)						
n	489	390	905	709	508	
me (\$10 ⁶)	1389	2380	2459	2850	2480	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the TSI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198506 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.37 [3.46]	0.34 [3.18]	0.41 [4.05]	0.41 [3.98]	0.48 [4.71]	0.48 [4.65]
Quintile	NYSE	EW	0.13 [2.19]	0.10 [1.72]	0.12 [2.09]	0.11 [1.94]	0.11 [1.90]	0.11 [1.80]
Quintile	Name	VW	0.26 [2.43]	0.21 [1.97]	0.27 [2.79]	0.27 [2.69]	0.37 [3.74]	0.36 [3.63]
Quintile	Cap	VW	0.34 [2.97]	0.28 [2.46]	0.35 [3.35]	0.34 [3.22]	0.47 [4.58]	0.46 [4.45]
Decile	NYSE	VW	0.38 [2.58]	0.38 [2.57]	0.45 [3.15]	0.40 [2.79]	0.53 [3.66]	0.49 [3.37]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.34 [3.17]	0.32 [2.98]	0.37 [3.72]	0.37 [3.69]	0.46 [4.55]	0.46 [4.48]
Quintile	NYSE	EW	-0.05 [-0.67]					
Quintile	Name	VW	0.23 [2.17]	0.19 [1.79]	0.24 [2.48]	0.24 [2.43]	0.35 [3.55]	0.34 [3.49]
Quintile	Cap	VW	0.31 [2.72]	0.25 [2.16]	0.30 [2.90]	0.30 [2.84]	0.43 [4.23]	0.43 [4.13]
Decile	NYSE	VW	0.34 [2.31]	0.35 [2.33]	0.40 [2.82]	0.37 [2.63]	0.50 [3.44]	0.47 [3.28]

Table 3: Conditional sort on size and TSI

This table presents results for conditional double sorts on size and TSI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on TSI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high TSI and short stocks with low TSI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198506 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	TSI Quintiles					TSI Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.82	0.57	0.24	0.50	0.99	0.17	0.16	0.17	0.14	0.09	0.07
		[2.81]	[1.79]	[0.57]	[1.39]	[3.15]	[1.35]	[1.24]	[1.36]	[1.08]	[0.66]	[0.52]
	(2)	0.74	0.85	0.48	0.53	0.96	0.22	0.19	0.22	0.21	0.25	0.25
		[2.64]	[3.12]	[1.31]	[1.74]	[3.23]	[2.27]	[1.95]	[2.40]	[2.27]	[2.63]	[2.53]
	(3)	0.76	0.77	0.69	0.78	0.84	0.08	0.07	0.12	0.10	0.15	0.14
	[2.70]	[3.06]	[2.05]	[2.78]	[3.00]	[0.70]	[0.56]	[1.04]	[0.92]	[1.29]	[1.18]	
(4)	0.89	0.78	0.79	0.88	0.94	0.05	0.00	0.08	0.04	0.22	0.18	
	[3.43]	[3.27]	[2.79]	[3.28]	[3.39]	[0.39]	[0.01]	[0.70]	[0.35]	[1.82]	[1.51]	
(5)	0.54	0.81	0.75	0.75	0.92	0.39	0.33	0.39	0.39	0.52	0.52	
	[2.36]	[3.98]	[3.58]	[3.29]	[3.77]	[3.05]	[2.54]	[3.20]	[3.16]	[4.32]	[4.26]	
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	TSI Quintiles					TSI Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	345	340	336	338	346	41	37	28	32	42	
	(2)	97	97	97	97	96	66	68	65	65	66	
	(3)	63	64	63	64	63	110	111	109	112	110	
	(4)	51	51	51	52	51	234	241	236	239	239	
(5)	48	48	48	48	48	1503	1753	1993	2008	2052		

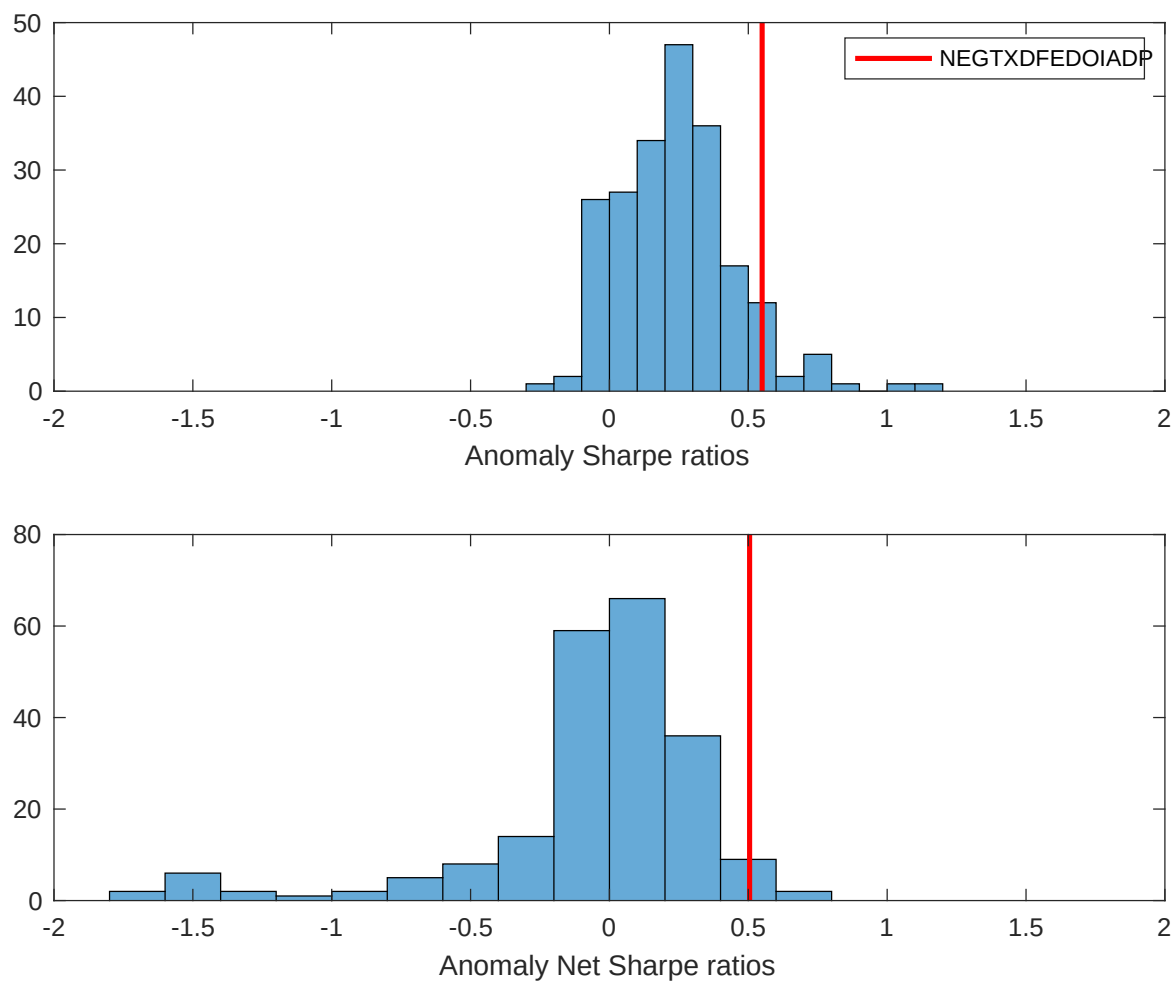


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the TSI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

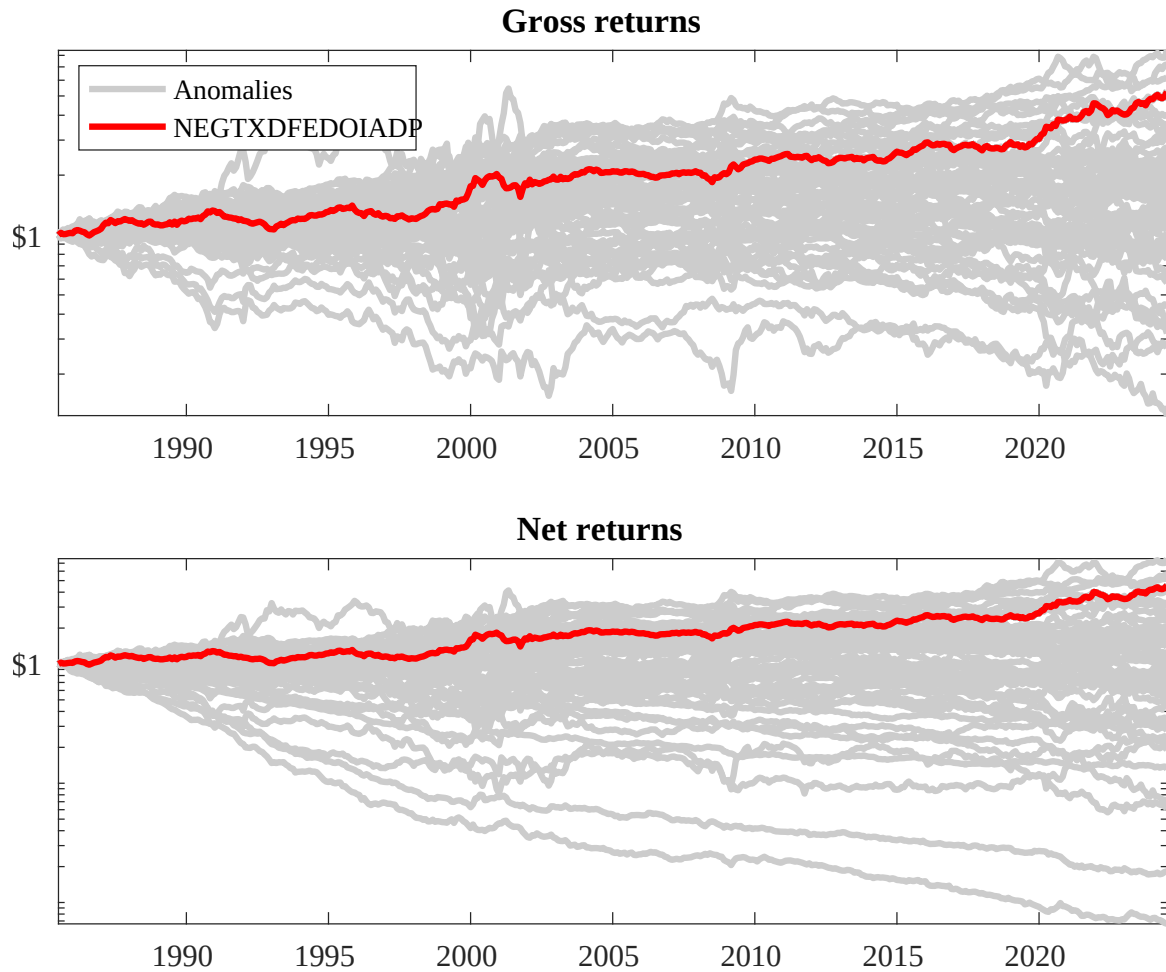
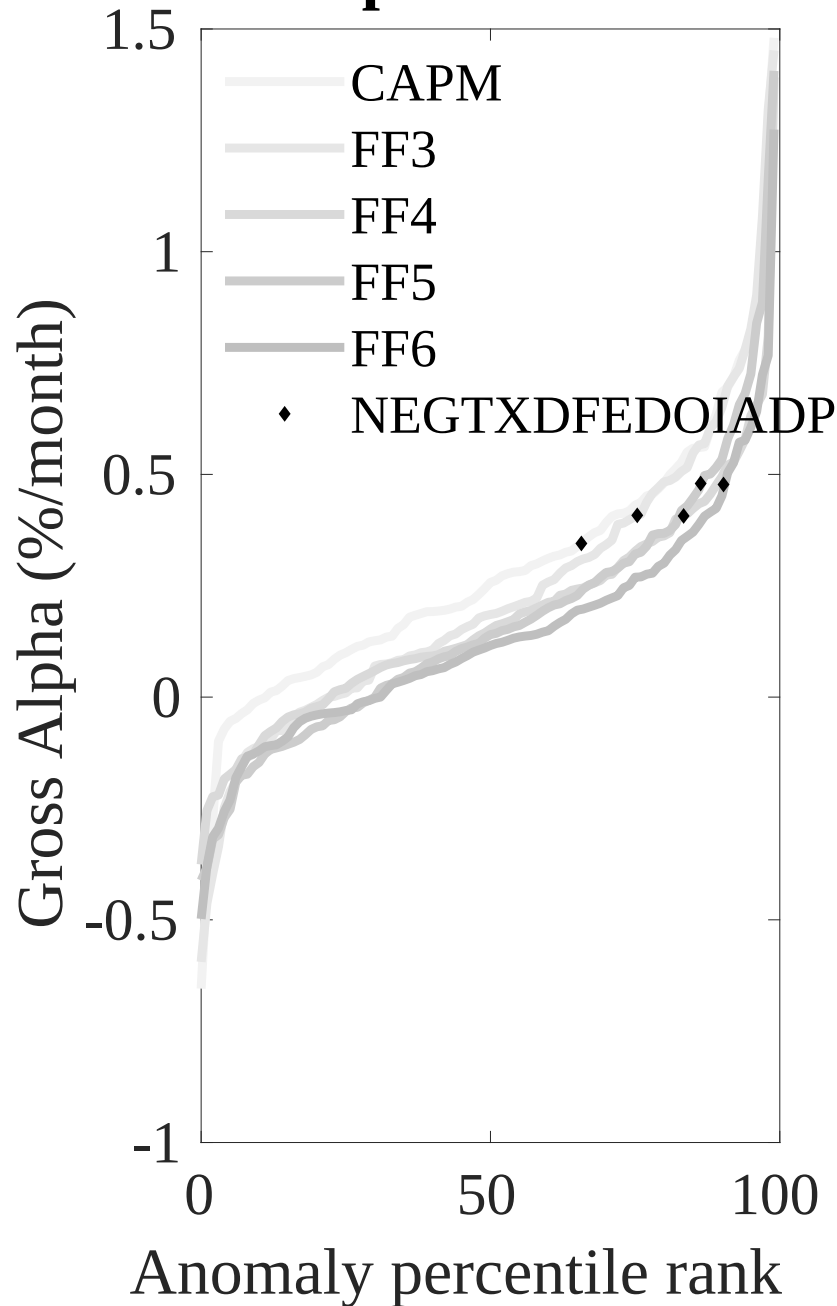


Figure 3: Dollar invested.

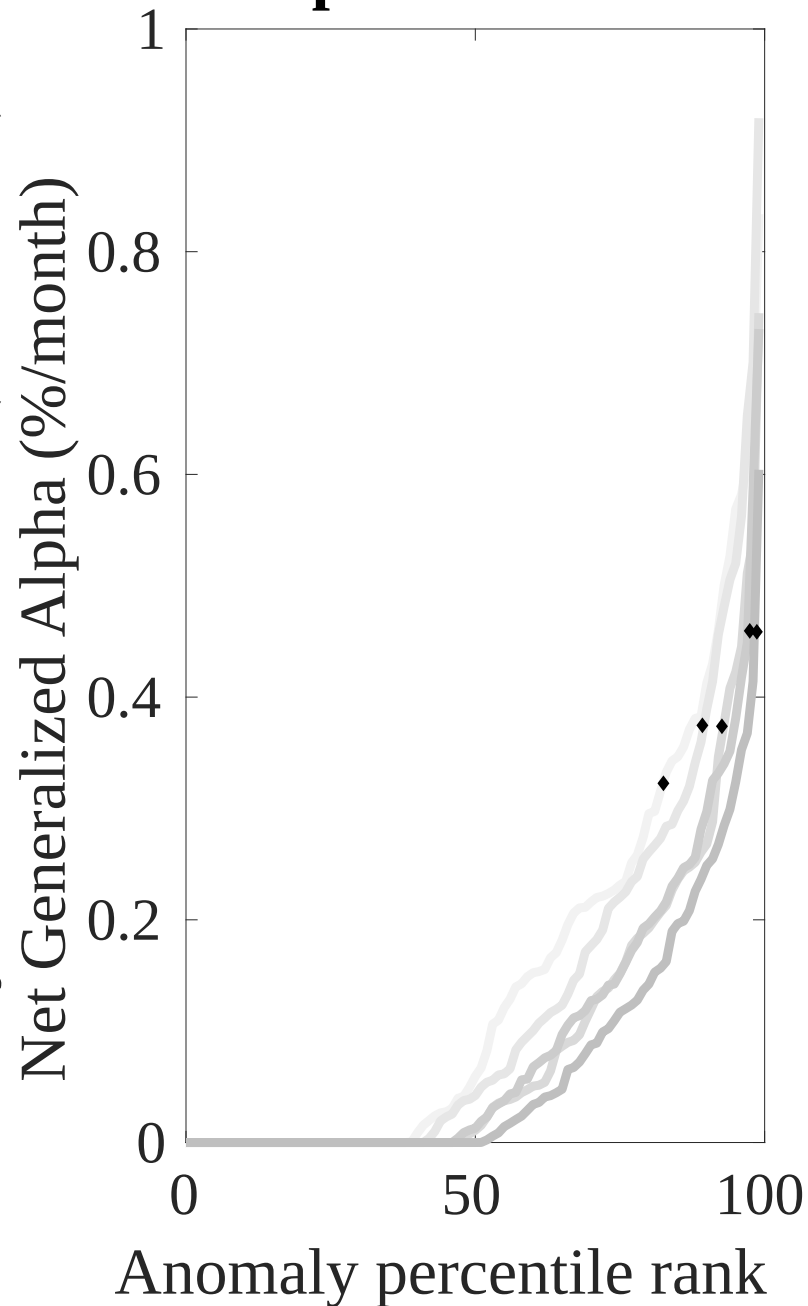
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the TSI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



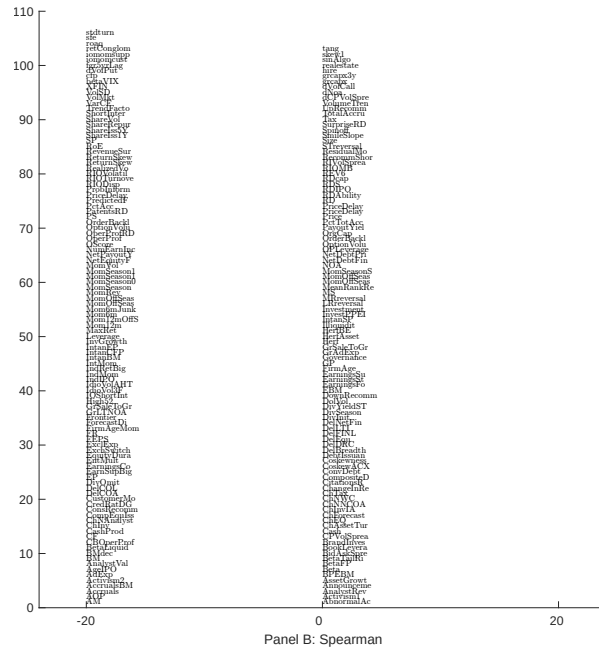
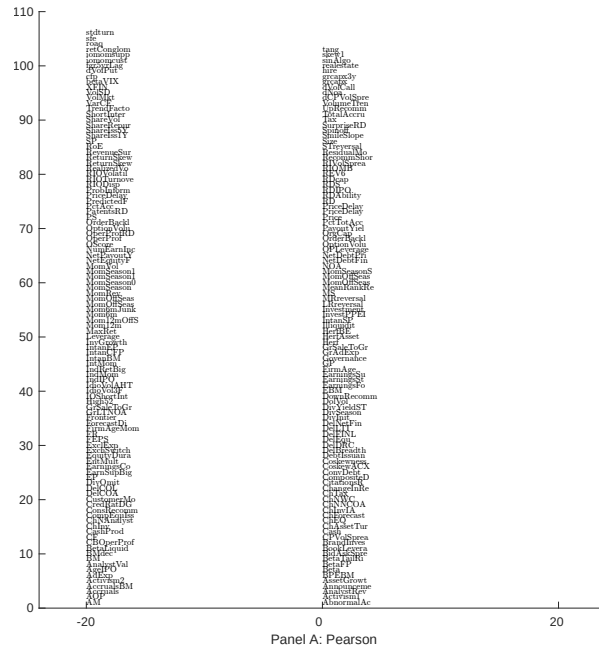


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with TSI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

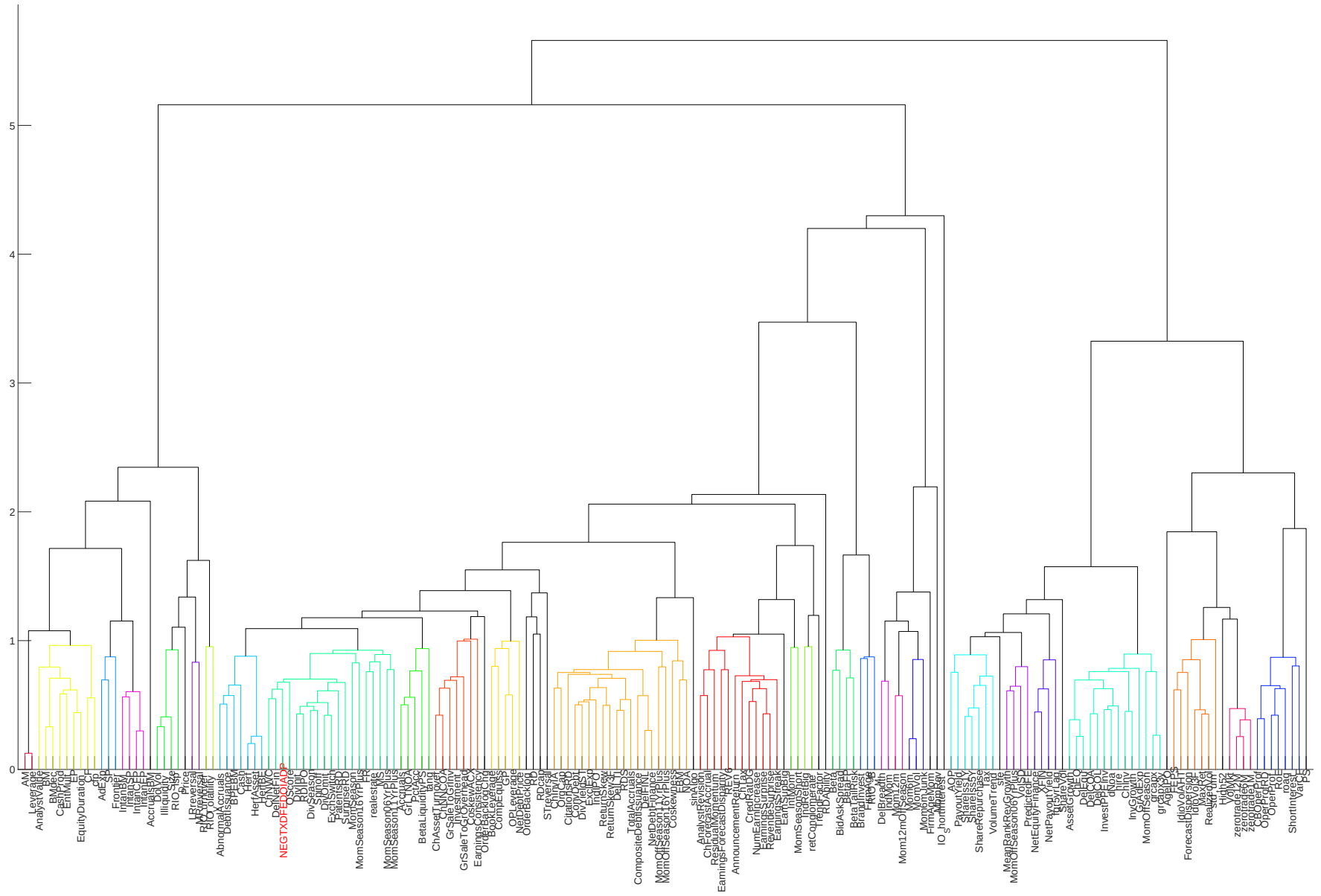


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

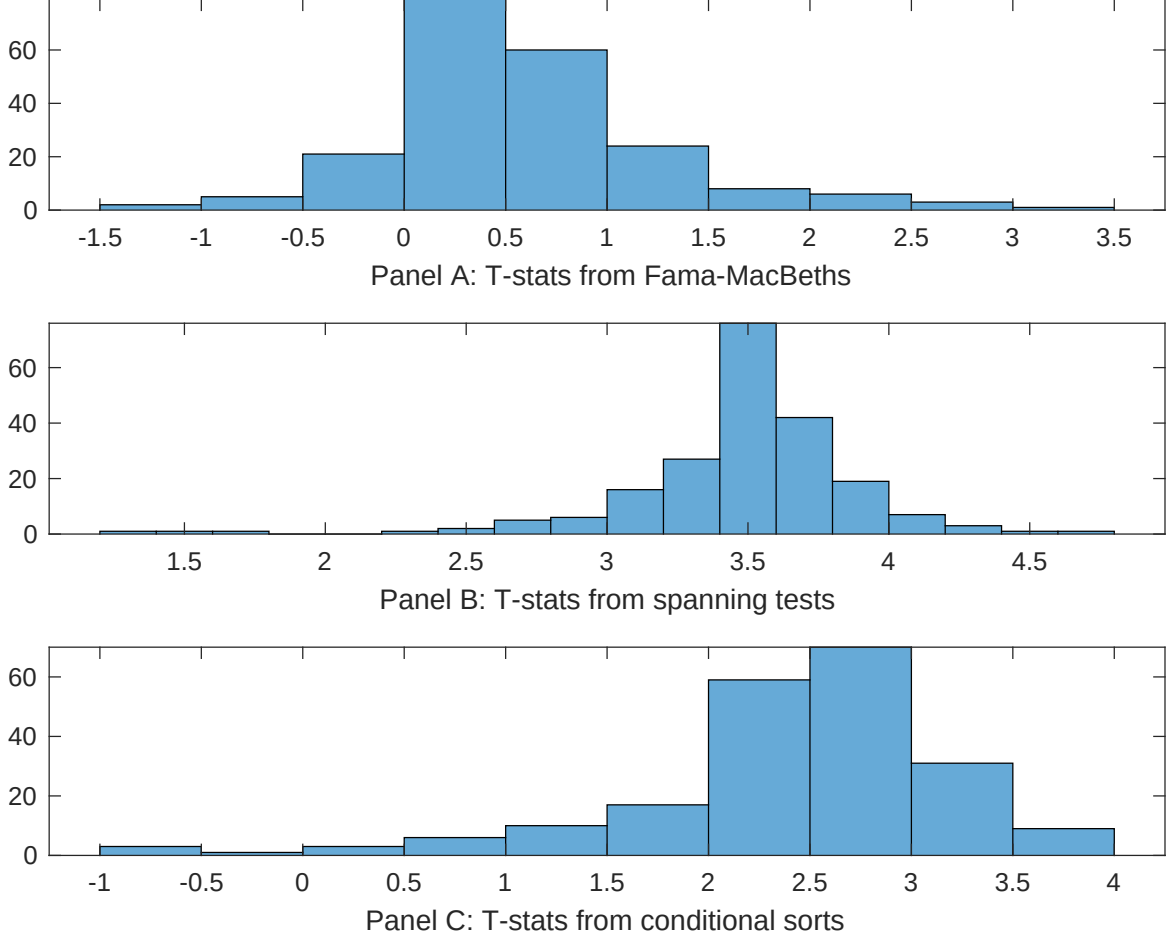


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of TSI conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TSI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TSI}TSI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TSI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on TSI. Stocks are finally grouped into five TSI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TSI trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on TSI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{TSI} TSI_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Cash to assets, Equity Duration, Cash flow to market, Book to market using December ME, Market leverage, Operating Cash flows to price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198506 to 202406.

Intercept	0.10 [3.69]	0.16 [6.32]	0.10 [3.50]	0.93 [3.04]	0.10 [3.51]	0.10 [3.46]	0.12 [4.81]
TSI	0.78 [1.35]	0.18 [0.31]	0.24 [0.43]	0.34 [0.59]	0.39 [0.69]	0.34 [0.59]	0.43 [0.77]
Anomaly 1	0.62 [1.56]						0.11 [3.13]
Anomaly 2		0.25 [2.66]					0.23 [3.41]
Anomaly 3			0.18 [0.58]				-0.23 [-0.98]
Anomaly 4				0.27 [3.22]			0.20 [2.84]
Anomaly 5					0.13 [0.61]		-0.25 [-1.32]
Anomaly 6						0.85 [2.95]	0.79 [4.36]
# months	463	468	463	463	463	463	463
$\bar{R}^2(\%)$	1	0	1	0	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the TSI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{TSI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Cash to assets, Equity Duration, Cash flow to market, Book to market using December ME, Market leverage, Operating Cash flows to price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198506 to 202406.

Intercept	0.28 [2.84]	0.47 [4.63]	0.48 [4.73]	0.46 [4.56]	0.47 [4.56]	0.51 [5.00]	0.29 [2.90]
Anomaly 1	33.90 [8.54]						30.37 [7.01]
Anomaly 2		-13.35 [-2.49]					-1.87 [-0.29]
Anomaly 3			-8.48 [-2.40]				2.14 [0.48]
Anomaly 4				-25.89 [-3.87]			-9.98 [-1.45]
Anomaly 5					-12.58 [-3.00]		-6.98 [-1.62]
Anomaly 6						-13.55 [-3.62]	-3.78 [-0.87]
mkt	-6.58 [-2.77]	-0.34 [-0.14]	-0.21 [-0.09]	-1.30 [-0.54]	1.38 [0.56]	0.27 [0.11]	-5.33 [-2.14]
smb	-6.87 [-2.00]	-5.10 [-1.33]	-7.16 [-1.95]	-0.88 [-0.22]	-7.09 [-1.94]	-5.87 [-1.60]	-3.35 [-0.86]
hml	-12.23 [-2.75]	-10.22 [-1.36]	-19.96 [-4.06]	1.74 [0.21]	-8.23 [-1.16]	-14.92 [-2.86]	9.45 [1.00]
rmw	-5.99 [-1.26]	-22.11 [-4.77]	-19.30 [-4.01]	-29.19 [-5.93]	-24.13 [-5.18]	-17.73 [-3.71]	-10.77 [-1.97]
cma	18.37 [2.92]	1.37 [0.20]	8.59 [1.29]	6.95 [1.07]	3.20 [0.49]	8.92 [1.36]	15.48 [2.28]
umd	0.42 [0.20]	0.48 [0.21]	-2.98 [-1.07]	1.58 [0.69]	-2.06 [-0.83]	-4.75 [-1.73]	-1.50 [-0.49]
# months	464	468	464	464	464	464	464
$\bar{R}^2(\%)$	29	19	18	20	19	19	29

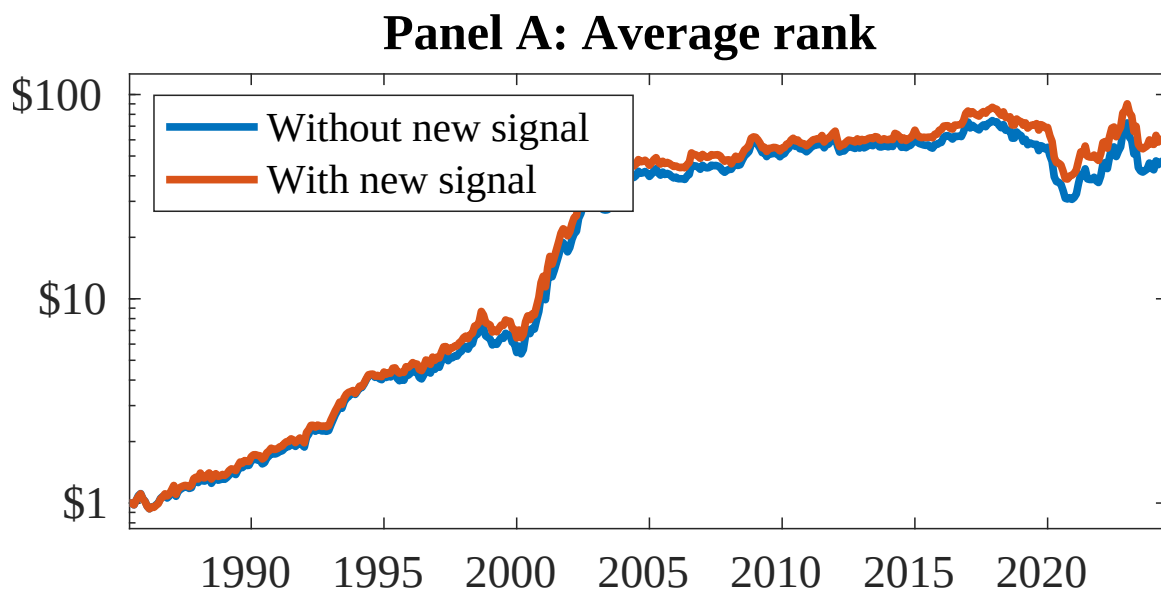


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as TSI. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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